

The friction cost method for measuring indirect costs of disease

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Abstract

A new approach for estimating the indirect costs of disease, which explicitly considers economic circumstances that limit production losses due to disease, is presented (the friction cost method). For the Netherlands the short-term friction costs in 1990 amount to 1.5–2.5% of net national income (NNI), depending on the extent to which short-term absence from work induces production loss and costs. The medium-term macro-economic consequences of absence from work and disability reduce NNI by an additional 0.8%. These estimates are considerably lower than estimates based on the traditional human capital approach, but they better reflect the economic impact of illness.

Keywords: Indirect costs; Absence from work; Friction costs; Cost of illness

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1. Introduction

The inclusion of indirect costs of disease in economic appraisal is subject to considerable debate and scepticism. The disagreement on the relevance of indirect costs resulting from reduced productivity can be illustrated by considering recent

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guidelines for economic evaluations in health care that are issued by Australian and Ontarian health authorities in the context of decisions on reimbursement of pharmaceutical products. The Australian guidelines suggest to exclude indirect costs, whereas the Ontarian seem to support inclusion (Commonwealth of Australia, 1990; Ontario Ministry of Health, 1991). Apart from ethical objections to any form of incorporating indirect costs in economic evaluation, criticism concentrates on the predominating estimation method: the human capital approach. Defining indirect costs as the value of production lost, more precisely as wealth lost, to society due to disease (and hence not as valuing lives per se), the human capital method estimates the *potential* production lost, which may overestimate the actual production lost to a considerable extent.

We introduced an alternative approach: the so-called "friction cost method", which takes into account several economic circumstances, that may reduce the estimated production losses substantially as compared with estimates based on the traditional human capital approach (Koopmanschap and Van Ineveld, 1992). In this paper we will develop this friction cost method further, focusing on the influence of the labour market situation on indirect costs, the impact of absence from work on labour productivity and the macro-economic consequences of absence from work and disability. The estimation of indirect costs takes place in two steps. First, the short-term effects of illness are considered by studying its possible consequences at the level of the firm. Second, medium-term effects occur because of changes in labour costs per unit of output, social insurance premiums and labour supply and these will influence national income, the most generally used yardstick for society's wealth, as well as other economic indicators such as (un)employment, inflation, the balance of payments and the government deficit.

As an illustration, total cost of absence from work, disability and mortality, respectively, will be estimated for the Netherlands in 1988 and 1990, using the friction cost method and the human capital approach. However, in this case estimates of total costs of disease represent the possible savings of eradicating disease and hence come down to estimating "the benefits of the unattainable" (Shiell et al., 1987). Therefore, the more relevant *consequences of attainable reductions in absence from work and disability* will be calculated. The impact of several assumptions will be investigated in a sensitivity analysis.

2. The friction cost method

2.1. Indirect costs disputed

Including indirect costs in economic appraisal of health care programs is not universally accepted. Opponents of inclusion state that health care programmes directed to working people always generate more gains in production than programmes for mentally handicapped or elderly. Even if allocation of resources

to individuals is not driven by such considerations, this could well lead to priority setting enlarging existing inequalities in health. This argument should be taken seriously. Nevertheless, production losses due to illness do influence the scarcity of resources and therefore the wealth of society. In this respect there is no difference in assessing direct and indirect costs; both should be incorporated in economic evaluations of health care programmes.

There may be a trade-off between equity and efficiency: society's wealth will indeed gain more from programmes directed to economically active people. Explicit estimates of the extent of the trade-off may be more helpful in decision making than using implicit notions. In health policy decisions, estimates of indirect costs should be weighed in conjunction with other criteria like direct costs, ethical considerations and equity.

The "costs" of pain, suffering, discomfort etc. are sometimes included in the concept of indirect costs. In this paper we focus on the indirect "economic" costs of illness¹.

2.2. *The human capital approach and the friction cost method*

Almost all studies estimating indirect costs use the human capital approach (Koopmanschap and Rutten, 1993). This method estimates the value of *potentially* lost production (or the potentially lost income) as a consequence of disease. We use the word *potentially* because in case of permanent disablement or premature death at a specific age the total productive value (or income) from that age until the age of retirement is counted as indirect costs. Many authors have suggested that the real production losses for society may be much smaller (Drummond, 1992; Gerard et al., 1989; Lindgren, 1981; Williams, 1985). For short-term absenteeism, work may be taken over by others, non-urgent work may be cancelled or made up for by the sick employee on his return to work. For long-term absences, work can be taken over by someone drawn from the ranks of the unemployed or by reallocating employees over jobs.

More systematically, if an employee gets ill, the following outcomes are possible for a firm's production and costs (assuming some form of social insurance for absence):

- (1) Both the level of production and costs are unaffected. This situation may occur if work can be made up for by the sick employee on his return to work or if internal labour reserves exist, allowing work to be taken over by colleagues without extra costs. The opportunity costs of internal labour reserves depend

¹ It seems more appropriate to consider these items as outcome variables and to quantify them in terms of reduced quality of life. The latter procedure reduces the risk of double counting (both as costs and health effects).

on the probability for the internal labour reserve to be gainfully employed elsewhere. If unemployment is well above the level of frictional unemployment, these costs are very low. However, the existence of permanent internal labour reserves raises labour costs, which may have medium-term macro-economic implications, as described below.

- (2) Production remains unchanged, but at higher costs, due to colleagues working overtime or hiring temporary workers, from a firm's own pool or from temporary agencies. In both cases the extra costs of maintaining production tend to be somewhat higher than average labour costs, as a result of higher wages paid for working overtime (reflecting the opportunity costs of leisure time) or the extra costs of using temporary agencies.
- (3) Production falls, while costs remain unchanged. The value of production lost is the relevant outcome.
- (4) Production falls, despite higher costs. The consequences are a mix of production loss and extra costs of permanent or temporary employees, which may be higher (or lower, although this is not to be expected) than the value of production of the sick employee.

In case of Situation 1, zero costs are incurred in the short run, whereas the medium-term consequences need to be analyzed, see below. Concerning the other three possible situations, the sum of production loss and extra costs varies from case to case, but on average it may well be approximated by the productive value of the sick employee during the period of absenteeism.

Because in Situation 1 no short-term costs prevail, we need to know its probability of occurrence. Exact empirical data on this probability are non-existent, but there is some material on the underlying determinants of the employment policies of firms.

First, it should be known how often work can be made up for by the sick employee on his return to work. This possibility seems to be restricted to short-term absence for jobs for which production can be postponed for a while and which output is not an indispensable input for other jobs. For example, a bus driver must be replaced in order to deliver the transportation services, whereas a sick health economist may make up for (most of) lost production a week later. Because we lack evidence on the magnitude of this phenomenon, we performed a sensitivity analysis regarding this issue.

The occurrence of Situation 1 also depends on the degree to which firms maintain internal labour reserves, which in turn depends on the relative costs and risks of keeping internal labour reserves versus flexible forms of labour provision, such as temporary workers, flexible contracts etc. The most important determinants for internal labour reserves versus flexible labour appear to be (Williamson, 1981; Osterman, 1987; Mitchell and Zaidi, 1990):

- the macro-economic situation and the state of the relevant product market. The more uncertainty exists, the greater the employer's desire to make labour costs variable and execute flexible labour policies;

- the situation on the labour market. The tighter the labour market, the more a firm wants to maintain internal labour reserves, because flexible labour may become very scarce and expensive;
- the technology of the production process. The more delicate and risky the production technology used, the more skilled employees are required and the more suitable internal employment is;
- the firm-specificity of employee's skills. In a downturn market employees with firm-specific skills will not be laid off soon, since training new employees if the market recovers is expensive;
- the predictability of the level of production (business cycle) and the possibility of production in advance. The less predictable production, the higher the costs of a permanent internal labour reserve. If stock production is not possible, as often in producing services, flexible labour may be very useful to produce the output required;
- the heterogeneity of labour. The less homogeneous labour is, the larger the total amount of internal reserve required and the more expensive it is as compared to flexible labour reserves;
- the predictability and variance in time of absence from work. If the absence pattern is unpredictable and clustered in time, flexible labour reserves seems more attractive;
- the degree of competition. The more competitive a market, the larger the propensity to choose for less expensive, flexible labour reserves;
- the institutional setting. A liberal labour market legislation facilitates flexible labour policies, for example by allowing immediate dismissal.

In summary, choosing the appropriate mix of internal and flexible labour reserves depends on the trade-off between three different goals of a firm: minimizing wages for any given level of output, maximizing predictability, i.e. the confidence of the availability of qualified labour at foreseeable prices, and flexibility with respect to staffing levels and abilities of the labour force (Osterman, 1987). Enhancing predictability always has a certain price in terms of higher labour costs and/or less flexibility and vice versa.

Empirical research for the United Kingdom, the Netherlands and the USA clearly shows that during the eighties the amount of flexible labour used has increased, even during the upturn of 1983–1990 (Hunter et al., 1993; OSA, 1987; Elfring and Kloosterman, 1989; Mitchell and Zaidi, 1990). In addition, Hunter et al. found that the firm's main motives for extending flexible labour are demands for short-term cover as well as deliberate responses to increased product market uncertainty. Although these results unambiguously indicate a growing role for flexible labour, they can not answer the question on the relative quantity of internal labour reserve versus the use flexible labour arrangements. Hence, we assume for the moment that all short-term absence spells induce indirect costs to some extent (for long-term absence see below). Consequently, the estimates of friction costs as described below should be viewed as *upper bound estimates* of

the short-term indirect costs of disease. The lower bound estimates are described in the sensitivity analysis.

Returning to the friction cost method, its basic idea is that the amount of production lost due to disease depends on the time-span organisations need to restore the initial production level. We assume that if unemployment, registered and hidden, is beyond the level of frictional unemployment, sick employees can be replaced, after a period necessary for adaptation. Frictional unemployment is an inevitable part of unemployment, since filling vacancies takes time and some qualitative discrepancies between labour demand and supply always prevail. Production losses are assumed to be confined to the period needed to replace a sick worker: the friction period. This situation currently applies to the Netherlands, where the level of registered and hidden unemployment is substantial. In 1988 and 1990 registered unemployment was 10.2% and 8.2% respectively (CPB, 1992a). In addition, several studies indicate that 15–20% of the persons receiving disability benefits should be considered as unemployed, thus adding another 3% unemployment (Roodenburg and Wong, 1985; Vrooman and De Kemp, 1990).

In order to calculate the indirect costs of disease, using the friction cost method, the following questions need to be answered:

- when does a friction period occur?
- how long does a friction period last?
- how can the indirect costs during a friction period be determined?
- how can the medium-term macro-economic consequences of illness be estimated?

2.3. Operationalization of concepts

The frequency of friction periods

The possible flows between healthy, absent from work, disabled and dead, which depend on the specific social security arrangements, determine when friction periods occur ² (see Fig. 1 for the Dutch case). In case of “immediate death” (Flow 1) and work absence (Flow 5) a friction period occurs. Premature death while one is absent from work (Flow 3) does not induce an additional friction period if the work absence is longer than the friction period.

Incidence of disability (Flow 4) and death for disabled (Flow 2) do not cause friction costs when, as is the case in the Netherlands, one acquires the status of disabled only after being absent from work for some time (one year in the Netherlands). The friction period then occurs in a previous year. Hence, in order to estimate the total number of friction periods one needs data on the frequency and

² Fig. 1 shows the situation for employees in the Netherlands, the majority of employment. For self-employed the situation is slightly different, which has been taken into account, see our previous article (Koopmanschap and Van Ineveld, 1992).

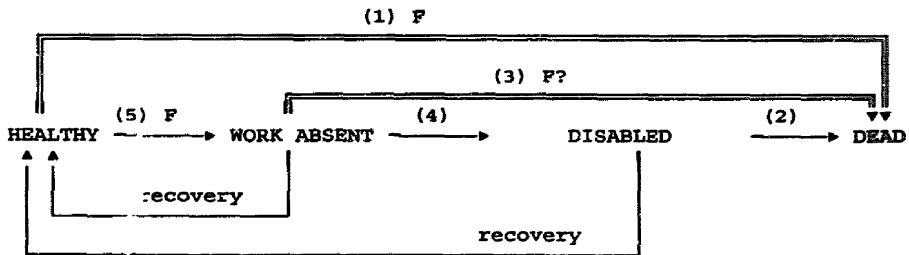


Fig. 1. Flows between healthy, absent from work, disabled or dead, and the relevance for friction periods (F).

length of absence spells, the incidence of disability and mortality, correcting for the number of deaths in the disabled population.

The length of the friction period

The length of the friction period is based on the average vacancy duration, which depends on the level of unemployment and on the efficiency of the labour market in matching labour demand and supply. Substantial unemployment in an economy as a whole does not rule out labour scarcity in specific parts of the labour market. Assuming a homogeneous labour market may result in underestimating indirect costs, and therefore one should consider at least different segments of the labour market separately, preferably according to education level. The education level has proved to be an important determinant of (un)employment (Teulings and Koopmanschap, 1989).

As a rule, the friction period is longer than the vacancy duration, since time may elapse between the emergence of production loss and the decision to create a vacancy. In addition, time passes between filling a vacancy and the first working day of the new employee, especially if he or she is already employed. Even filling a vacancy by an employee from the same firm takes time. We take both circumstances into account in estimating the length of the friction period.

Absence and productivity

As noted above, short-term absence may lead to production loss and/or extra costs to continue production, but the extent to which this occurs depends on various circumstances as discussed above. Complete insight in the consequences of short-term absence for indirect costs requires more detailed study on the level of the firm. However, generally speaking, absence from work reduces the effective labour time. Numerous studies have demonstrated that a reduction of annual labour time causes a less than proportional decrease in labour productivity per year. The estimated elasticity for annual labour time versus labour productivity is between 0.6 and 0.9. (CPB, 1987; De Koning and Tuyl, 1984; WRR, 1977). In our main variant we assume the elasticity to be 0.8.

It should be noted however, that because these elasticities were estimated on the level of the firm, they inevitably reflect the composite result of two possible components: the diminishing returns on labour and a possible reduction of the internal labour reserve. On the level of the firm, these two components can not be disentangled.

The valuation of lost production

We assume that the production level is restored after (a part of) the friction period. The actual indirect costs of disease consist of: the value of production lost and/or the extra costs to maintain production and, if an employee is to be replaced permanently, the costs of filling a vacancy and training new personnel. We assume the costs to be 80% (the aforementioned 0.8 elasticity of labour time versus productivity) of the average value of production per employee, by age-group, sex and education level. Costs of absence from work shorter than the friction period are calculated as being 80% of the production value during the period of absence. Costs of absence spells longer than the friction period are equal to the costs of a friction period, since we assume replacement after completion of the friction period. The production value per time period *per employee* is multiplied with the length of friction periods due to absence from work and incidence of disability. Because we assume equal mortality rates for employed and non-employed, the average production value *per person* (the production value per employee, corrected for participation in paid labour) is used to calculate the indirect costs of mortality before age 65.

Macro-economic consequences

The friction cost method concerns the short-term indirect costs of disease. In addition, absence and disability may also have medium-term macro-economic consequences. Absence from work, reducing labour productivity, directly affects the labour costs per unit of output, which is an important determinant of competitiveness on the world market. This may change the production of firms in the "exposed sector" of the economy: exporting firms as well as firms producing for the domestic market facing foreign competitors. Consequently, national income and other macro-economic indicators may be affected.

The amount of insurance premiums related to absence has no separate economic impact in the Netherlands. A change in absence premiums paid only causes a redistribution between wages and sickness benefits, total labour costs per worker remaining equal.

Although disability benefits themselves do not represent indirect costs, as they are income transfers, they may have economic consequences in two ways. First, insurance premiums influence the labour costs per worker and per unit of output, affecting the performance of the exposed sector of the economy. Secondly, insurance premiums affect the net disposable income of employees and employers, influencing private consumption, domestic production and national income. The

extent of this influence depends on the possibilities for employees and employers to pass on changes in premiums to each other.

Furthermore, disability may have other economic consequences, in addition to the “premium-route” just described. Reducing the incidence of disability will raise labour supply and unemployment. High unemployment has a negative medium-term impact on the average wage level and thus on labour costs, improving the competitiveness of the exposed sector. On the other hand, the high prevalence of disability in the Netherlands has undoubtedly raised the productivity of the labour force, because in the Netherlands less productive workers run a higher risk of becoming disabled (Aarts and De Jong, 1990). Policies, effective in lowering the incidence of disability, may have a small negative effect on average labour productivity.

These macro-economic consequences of absence and disability can only be estimated by using a macro-econometric model, allowing for all interactions and time lags involved.

2.4. Data and methods for the Netherlands

Frequency of friction periods. We used national, aggregated data on the frequency and length of absence from work, covering 50% of employment. We corrected for the fact that the absence rate for persons for whom we lack exact data is somewhat lower (CBS, 1989,1991). Data on incidence of disability and mortality by age and sex were available, covering the entire population (CBS, 1989a,1991a; GMD, 1989,1991).

The relation between absence and education level (and age and sex) for 1988 was derived from a sample of 25,000 Dutch workers (NIA, 1993). For the risk of becoming disabled by education level, we used the estimates of Bloemhoff and De Winter (1991). Mortality rates for men by education level for the Netherlands were estimated by Doornbos and Kromhout (1990) and applied to men and women.

Length of the friction period. We split up the labour market into 5 education levels. This reduced heterogeneity considerably, because in the Netherlands unemployment is clearly higher for lower educated (CBS, 1991b). Following Van Ours’ model (Van Ours and Ridder, 1991), we estimated the completed vacancy durations by education level for 1988 and 1990, based on quarterly data on the uncompleted vacancy durations and the number of vacancies, from a large sample of Dutch firms (CBS, 1988,1990), adding an additional period to allow for time lags indicated in Section 2.3 above.

Absence and productivity. The elasticity for annual labour time versus labour productivity, found for the Netherlands is between 0.6 and 0.9 (WRR, 1977; De Koning and Tuyl, 1984; CPB, 1987). We assume the elasticity to be 0.8 (De Koning and Tuyl, 1984), performing sensitivity analysis on this parameter.

Valuation of lost production. The average value of production per labour-year was approximated by combining the net national product and employment in labour years in 1988 and 1990 (CPB, 1992a). This production value was made age-, sex- and education level-specific using data on the relative gross labour incomes per labour-year by age, sex and education level (CBS, 1989b, 1991c; LTD, 1993). We assumed a linear relationship between gross income and value added per labour year. Data on participation in full-time and part-time labour, based on the annual Dutch labour force sample (CBS, 1989c, 1991b) were used to calculate the average production value *per employee* and *per person* (age-, sex- and education-specific).

The *macro-economic consequences* of absence and disability were calculated, using the quarterly macro-econometric sectoral model FKSEC for the Netherlands, developed by the Dutch Central Planning Bureau (CPB, 1992b). This general equilibrium model has been chosen after careful comparison of several macro-econometric models on their overall performance as their suitability for estimating the macro-economic consequences of disease. The model FKSEC is the youngest member of a family of well known Dutch macro-econometric models and is currently the most frequently used in advising the Dutch government and parliament on short- and medium-term economic development.

It explicitly models the production process on the sectoral level (6 sectors), taking account of the differences between sectors in labour intensity, in the degree of substitutability between capital and labour and in the speed with which different sectors adjust to shocks in demand. In the capital-intensive exposed sector, the production capacity is based on a vintage production function. Vintages of capital goods for which the labour productivity is lower than the real wage have a negative return and are scrapped. Employment in the exposed sector is related to labour demand at full capacity and the utilization rate of capital. Because in the sheltered sector and other sectors labour is the main factor of production, capacity is based on the current labour force and the lagged trend in labour productivity. The employment level is determined by production (given production capacity), real labour costs, the ratio between the minimum and the average wage, working time and labour augmenting technical progress.

A historical simulation for the period 1979–1988 was carried out with the model, showing that FKSEC is capable of reproducing the most crucial developments in the eighties, such as the decrease in export growth, investments and production in the beginning of the decade and the acceleration afterwards (CPB, 1992b). An extensive survey of the CPB macro-economic models is given by Bodkin et al. (1991).

The model does not have a separate variable for absence from work, but absence from work primarily influences labour time per year and hence labour productivity per worker. A change in the number of disabled principally affects labour supply, social insurance premiums and labour productivity. The model

simulates the rest of the consecutive chain of events including changes in labour costs, production, consumption, exports, national income etc.

The macro-economic consequences of disease are calculated as differences between the simulation runs and the base run. The latter reflects the expected economic development, without serious changes in the development of absence and disability. The simulation period is 1991–1998, allowing all possible time-lags to effectuate. In the simulation runs changes in absence and disability are assumed to be structural.

For the *main variant* in the following results we have split the labour market into five education levels and assume the elasticity of labour time versus production to be 0.8 (De Koning and Tuyl, 1984). Absence from work, incidence of disability and mortality are also related to sex, age and education level.

3. Results

3.1. Short-term friction costs

The average friction period was estimated to be 2.8 months in 1988 and 3.2 months in 1990, including the vacancy duration and 3–5 weeks to allow for the time lags mentioned (Table 1). In both years the friction period is considerably longer for higher education levels, reflecting higher unemployment for lower educated and a longer period needed for hiring higher educated workers. In 1990 friction periods converged, as the labour market for higher educated became less tight, while at the same time lower and intermediary educated technicians became relatively more scarce.

The average production loss per employee, per friction period is higher for men than for women and increases sharply with education level and with age, levelling off at age 50 (see Appendix). This reflects the strong positive relation between hourly wages and both age and education level and the substantial amount of part-time work performed by women in the Netherlands.

Absence from work is strongly related with education: absence rates for the lowest educated are about 4 times as high as for the highest education level. Absence increases with age and is higher for women under 45 compared to men,

Table 1

Estimated length of friction periods for the Netherlands in months, by education level, for 1988 and 1990

Year	Overall	Education level				
		Basic	Extended basic	Intermediary	Higher vocational	University
1988	2.8	2.2	2.5	2.7	3.5	3.8
1990	3.2	2.8	3.0	3.3	3.4	3.5

Table 2

Indirect costs of disease in the Netherlands for 1988 and 1990 (main variant) in billions of guilders (between parentheses as % of net national income)

Cost category	Friction costs 1988	Human capital costs 1988	Friction costs 1990
Absence from work	9.2	23.8	11.6
Disability	0.15	49.1	0.2
Mortality	0.15	8.0	0.2
Total indirect costs	9.5 (2.1%)	80.9 (18%)	12.0 (2.6%)
<i>without education split</i>	9.3		11.6

partly due to pregnancy. The lower educated also run a higher risk of becoming disabled: the relative risk of the lowest versus the highest education level is 2.3, after correction for age. The relative risk of dying before age 65 for the lowest educated is 1.5.

The friction costs of disease for the main variant (assuming a heterogeneous labour market and labour time elasticity of production = 0.8) are shown in Table 2. The friction costs amount to 9.5 billion guilders in 1988 and 12 billion guilders in 1990, about 2.5% of net national income. About half of the cost increase between 1988 and 1990 is due to employment growth (raising the absolute number of absence days), productivity growth and inflation. The other half is the result of (on average) longer friction periods, mirroring a more tight labour market. The costs of absence from work predominate, since we assume production losses to be confined to the friction period and consequently disability and mortality hardly contribute to the costs. The distinction by education level has a small upward effect on costs. The higher costs per hour and longer friction period for higher educated are almost entirely compensated by their considerably lower rate of absence.

Table 2 also presents the indirect costs for 1988, according to the human capital approach, which amount to more than 80 billion guilders, 18% of net national income. The huge difference compared to the friction costs is due to the cost of disability and mortality, which are assumed to cause production losses in the long term. Costs of disability are very large, because the average duration of disability in the Netherlands is 15 years.

The costs of absence from work differ less dramatically. As the friction cost method assumes production loss to be limited to the short run, the friction costs for diseases that cause mainly disability and mortality are much lower than according to the human capital approach. For diseases entailing short-term absence, the difference is much smaller.

3.2. Medium-term macro-economic consequences

The medium-term macro-economic consequences of absence and disability are presented in Table 3, as structural differences in 1998 and later years, between the

Table 3

Macro-economic consequences of eliminating absence from work and disability respectively, during 1991–1998, in % of net national income and other macro-economic indicators (calculated as differences in 1998 between the simulations and the base run)

	% NNI	Employment × 1000	Production	Export	Balance of payments % of NNI	Inflation	Govern. budget % of NNI
I Absence from work	+ 0.3%	– 112	+ 0.9%	+ 2.0%	+ 0.1	– 0.7%	– 0.2%
II Disability total effect	+ 0.49%	+ 345	+ 2.2%	+ 4.0%	+ 0.8%	– 4.2%	– 1.5%
Ila Disability premium-effect	+ 0.07%	+ 71	+ 0.4%	+ 1.0%	+ 0.5	– 0.9%	+ 0.6
Ilb Disability labour supply effect	+ 0.42%	+ 274	+ 1.9%	+ 3.1%	+ 0.3%	– 3.3%	– 2.1%

specific variants and the base run. The average Dutch absence rate is 8%, so we increased working time by 9% ($= \{100 - 92\}/92$) during 1991–1998 to illustrate the impact of eliminating absence from work. In 1998, overall labour productivity will be 3.3% higher than in the base run: a 5% increase in the exposed and construction sector and 2% in the sheltered sector. The production of the exposed sector will increase by 1.6%. In the sheltered sector and the construction sector the production increase is only 0.6%, since it is limited by domestic demand. As a consequence, employment falls, raising unemployment.

Labour costs per unit of output decrease, causing an increase in export (+ 2%) and overall production (+ 0.9%) from 1994 onwards. The increased unemployment will push down wages, so private consumption slightly falls. The decrease in consumption occurs almost immediately, whereas the increase in production and export is lagged. As a consequence, it takes 4 years for the national income to increase as compared to the base run. After 8 years the structural increment in national income is 0.3%, which is considerably smaller than the increase in export.

Assuming no incidence of disability implies that each year during 1991–1997 labour supply increases by 116,000 persons, who would otherwise have left the labour force. As a result, disability premiums and the corresponding cash benefits decrease by 3.1 billion guilders per year. The second row of Table 3 shows the total consequences of disability. Labour costs per worker decrease substantially (– 16% in 1998), due to lower premiums and increased labour supply (the “Phillips curve-effect”). Employment grows considerably, but it can only absorb one third of the extra labour supply, so unemployment rises. Production (+ 2.2%) and export increase (+ 4%), whereas private consumption decreases (– 1.7%), due to lower wage rates.

After 5 years, national income will be higher as compared to the base run, in 1998 the difference is 0.49%. The last two rows of Table 3 show that the increase in labour supply is responsible for the lion's share of the effects. The impact of the lower disability premiums is relatively small, due to the trade-off between disability benefits and unemployment benefits. Since the total effect of disability is

Table 4

Change in friction costs and macro-economic consequences of three scenarios concerning reduction of absence from work and incidence of disability. Friction costs in billions of guilders (1990) and in parentheses as percentages of net national income (1990). Macro-economic consequences in percentages of NNI in 1998

Scenarios		Change in friction costs in bln dfl	Macro-economic consequences in % of NNI
No 1:	Absence – 2%	– 2.3 (0.5%)	0.08%
No 2:	Disability incidence – 24000	– 0.03	0.1%
No 3:	Scenario 1 and 2	– 2.33 (0.5%)	0.18%

about equal to the sum of both effects, interaction effects appear not to be significant.

3.3. Scenarios

Table 4 shows the friction costs and the macro-economic consequences of three attainable scenarios regarding reduction in absence from work and disability in the Netherlands. These scenarios appear to be attainable, witnessing recent trends towards more prevention and closer monitoring of absence from work in the Netherlands. Furthermore, there is scope for reduction as the level of absence from work in Belgium and Germany is lower than in the Netherlands (Prins, 1990).

Reducing absence from work by 2% (Scenario 1), from 8% to 6% of labour time, will reduce friction costs by 2.3 billion guilders, about 0.5% of net national income. The medium-term economic effects induce another 0.08% annual increase in NNI. Reducing the incidence of disability by 20% (Scenario 2) hardly saves any friction costs, but the macro-economic consequences are more important: 0.1% of NNI. Reducing both absence and disability (Scenario 3) will save 2.3 billion guilders friction costs and 0.18% of net national income in the medium term. The effect of scenario 3 turns out to be more or less equal to the sum of the effects of Scenario 1 and 2. The other macro-economic indicators are affected in the same direction as shown in Table 3.

3.4. Sensitivity analysis

The results of the sensitivity analysis are presented in Table 5. The first two options represent the lower bound estimates for the friction costs of absence from work.

As we lack detailed data on the consequences of short-term absence, we assumed quite radically that half of the absence spells shorter than one week will not cause indirect costs, because work can be cancelled or postponed and made up for later. In this situation, the friction costs of absence from work would amount to 10.2 billion guilders in 1990, which is only 12% lower as compared to the main variant.

Table 5

Sensitivity analysis: indirect costs of absence from work (excluding medium term macro-economic consequences) in billions of Dutch guilders in 1990. In parentheses the percentage differences as compared to the main variant

Type of sensitivity analysis	Indirect costs of absence (% diff.)
Postponement of work	10.2 (– 12%)
Elasticity of production 0.5	7.3 (– 37%)
Friction period 6 months	14.7 (+ 27%)
Human capital approach	23.8 (+ 105%)

If we alternatively assume the elasticity of labour time versus production to be very low: 0.5 instead of 0.8, (implicitly assuming an important role of internal labour reserves) the friction costs of absence are substantially lower: 7.3 billion guilders, about 1.5% of NNI, a 37% decrease as compared to the main variant. Hence, the friction costs are very sensitive to the level of this elasticity.

In order to show the relation between the labour market situation and friction costs, we assumed a very tight labour market, entailing a long friction period, say 6 months. The friction costs amount to 14.7 billion guilders, a 27% increase as compared to the main variant. Indirect costs of absence from work based on the human capital approach amount to 23.8 billion guilders, which is considerably more than both the main variant and the variant with the 6 months friction period. The latter difference stems from the fact that in the Netherlands the proportion of long-term absence is quite large. More than 20% of all absence days are accounted for by persons who stay the maximum of one year in the absence insurance scheme.

4. Discussion

The friction costs and the macro-economic consequences of absence from work do not overlap, but are complementary. The friction costs are restricted to the short-term consequences of disease. After completion of the friction period, no costs are considered. The influence of absence on labour time, productivity, labour costs and production (especially of the exposed sector) is a medium-term economic process, which takes several years to occur. Furthermore, in the macro-economic analysis, the negative price elasticity of demand ensures that lower production costs stimulate foreign demand. The friction cost method does not take this interaction of price and demand into account.

Micro- and macro-economic effects are complementary in describing the comprehensive effects of absence from work and disability. The macro-economic situation influences the length of the friction period and the relative importance of internal labour reserves, both determinants of friction costs. On the other hand, the

internal labour reserves influence labour costs, which has clear macro-economic implications in the medium term.

The exact consequences of short-term absence from work for labour productivity and production need to be analyzed in more detail on the level of the firm. This analysis should deal with the main issues discussed: the possibilities of postponing work and making up for it later, the nature and extent of internal labour reserves versus flexible labour and the diminishing returns to labour on the level of the individual employee.

If unemployment is very low, the friction period is relatively long. By then the vacancy duration may become a less accurate proxy of labour scarcity, as some firms will give up hiring personnel. Consequently, the registered vacancy duration may underestimate the true vacancy length and the friction costs may underestimate the value of production losses. If an absolute labour shortage would occur on a particular part of the labour market, the friction period could become as long as it takes to educate new employees appropriately. For the exposed sector of the economy this situation could mean a loss in market share and output.

Facing a tight labour market, firms tend to maintain more internal labour reserves, but at the same time hiring employees is more difficult, mitigating this tendency.

We suggest that the true indirect costs of disease are within the range between the minimum estimates as provided in the sensitivity analysis and the estimates of the main variant. The minimum estimates are very conservative estimates of indirect costs, as these incorporate quite radical assumptions concerning the opportunity for postponing work and the impact of absence on productivity, respectively. This range of estimates is more useful in presenting the economic consequences of disease in the real world as compared to human capital estimates, showing the potential consequences of disease for an economy in a state of full employment equilibrium only found in textbooks.

The level of friction costs in other countries depends on the labour market and the relation between absence and education. The macro-economic consequences for other countries may be somewhat smaller. The Netherlands is a small open economy with a large exposed sector, hence, economic growth is very sensitive to changes in labour costs. However, not only the exposed sector is affected but also firms producing intermediary supplies. Furthermore, the rapid international economic integration will intensify international competition for firms producing for the domestic market, enlarging the importance of the exposed sector worldwide. The Dutch level of disability is relatively high, so reducing disability may have relatively large macro-economic effects as compared to other countries.

Although our analysis is age-, sex- and education-specific, we used fairly aggregate data which may not reflect the indirect costs for specific groups of patients. For estimating indirect costs on the individual level patient-questionnaires seem more appropriate. The latter may also be useful in estimating productivity loss without being absent from work (e.g. for ulcer or migraine) which hardly have

been studied thus far. Production losses without absence may increase if absence from work is reduced without a decrease in the prevalence of the underlying disease.

Estimates of indirect costs should incorporate production loss in relation to unpaid labour to prevent adverse equity implications. However, data on work absence and disability related to unpaid labour are extremely scarce. A combination of data from time-surveys and patient questionnaires, administered by those carrying out unpaid work may provide reliable data.

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Appendix 1

Average production value per employee per friction period, by age, sex and education level for the Netherlands 1990, in thousands of Dutch guilders

Age	Education				
	Basic	Extended basic	Intermediary	Higher vocational	University
<i>Men</i>					
15-19	4.6	5.1	7.1	Not applicable	Not applicable
20-24	10.8	11.9	15.0	18.6	20.6
25-29	13.3	15.9	20.3	23.5	26.3
30-34	14.8	18.0	24.3	31.7	35.6
35-39	14.4	17.9	27.9	34.7	56.9
40-44	15.2	18.7	29.3	41.6	65.9
45-49	15.9	19.3	31.5	42.1	63.2
50-54	15.5	18.6	32.2	44.1	58.8
55-59	14.6	18.4	28.9	40.8	59.3
60-64	13.0	15.2	26.8	36.6	53.3
<i>Women</i>					
15-19	3.5	3.7	6.3	Not applicable	Not applicable
20-24	7.6	9.4	12.6	17.1	19.7
25-29	9.1	11.2	15.6	20.1	23.6
30-34	7.8	10.1	15.4	20.4	24.0
35-39	8.3	10.5	17.1	22.1	32.8
40-44	7.7	9.6	16.4	24.3	33.6
45-49	7.8	10.2	17.8	26.5	38.0
50-54	7.4	10.1	18.6	27.1	37.9
55-59	6.7	9.0	16.8	24.6	33.5
60-64	6.2	8.2	15.8	23.1	31.4

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